United States Patent Application Publication	20250283541
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	LIN; Bingrong et al.

SPRING-LOADED MULTI-CHANNEL ELECTROMAGNETIC VALVE

Abstract

Disclosed is a spring-loaded multi-channel electromagnetic valve, including a valve core assembly, a valve seat assembly, a sealing unit, and a power unit. The valve seat assembly includes a valve seat and a valve cover. The valve core assembly includes a valve core body and a stop member. The stop member is connected to the valve core body. The sealing unit includes a sealing ring and an elastic member. The sealing ring is provided with a plurality of through holes. The sealing ring is connected to the valve seat, and the elastic member is connected to the valve core body and the stop member. The power unit is connected to the stop member and drives the valve core body to rotate.

Inventors: **LIN; Bingrong (Ningbo, CN), XU; Junbo (Ningbo, CN), DAI; Haijiang (Ningbo, CN), WU; Chunming (Ningbo, CN)**

Applicant: **Zhejiang ZEEKR Intelligent Technology Co., Ltd.** (Ningbo, CN); **Zhejiang Geely Holding Group Co., Ltd.** (Hangzhou, CN)

Family ID: **86962980**

Assignee: **Zhejiang ZEEKR Intelligent Technology Co., Ltd. (Ningbo, CN); Zhejiang Geely Holding Group Co., Ltd. (Hangzhou, CN)**

Appl. No.: **19/217758**

Filed: **May 23, 2025**

Foreign Application Priority Data

CN	202310357138.6	Mar. 30, 2023

Related U.S. Application Data

parent WO continuation PCT/CN2024/072713 20240117 PENDING child US 19217758

Publication Classification

Int. Cl.: **F16K11/074** (20060101); **B60H1/00** (20060101); **F16K27/04** (20060101)

U.S. Cl.:

CPC **F16K11/074** (20130101); **F16K27/045** (20130101); **F16K27/048** (20130101); B60H1/00485 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International Application No. PCT/CN2024/072713, filed on Jan. 17, 2024, which claims the benefit of priority to Chinese Application No. 202310357138.6, filed on Mar. 30, 2023, both of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD
[0002] The present disclosure relates to the field of electromagnetic valves for new energy vehicles, in particular to a spring-loaded multi-channel electromagnetic valve.
BACKGROUND
[0003] With the continuous promotion and development of new energy vehicles, pure electric vehicles have gained increasing popularity.
[0004] A temperature control system inside the pure electric vehicles is interconnected through electromagnetic valves to achieve temperature control for the entire vehicle. The electromagnetic valves include multi-channel electromagnetic valves and integral electromagnetic valves. The multi-channel electromagnetic valves require connection via pipelines, while each integral electromagnetic valve includes a valve core and a valve body, forming a passage for a cooling liquid through rotational cooperation of the valve core and the valve body.
[0005] However, during long-term use, these two types of electromagnetic valves are prone to pipeline aging and wear between the valve core and the valve body, which can lead to a reduction in the overall sealing performance of the electromagnetic valves, thereby affecting the temperature control effect of the vehicle.
SUMMARY
[0006] The present disclosure provides a spring-loaded multi-channel electromagnetic valve, including: a valve core assembly, a valve seat assembly, a sealing unit, and a power unit; where the valve seat assembly includes a valve seat and a valve cover, the valve seat is a rotary member, two ends of the valve seat are respectively provided with two openings to form a first accommodating cavity, the valve cover covers one of the two openings, and each of the valve core assembly and the sealing unit is located in the first accommodating cavity; the valve core assembly includes a valve core body and a stop member, a shape of an outer wall of the valve core body is consistent with a shape of the opening, the valve core body is provided with a plurality of channels extending along an axial direction of the valve seat, the plurality of channels are distributed at intervals along a circumferential direction of the valve core body; the stop member is connected to the valve core body; the sealing unit includes a sealing ring and an elastic member, a shape of the sealing ring matches the shape of the opening of the valve seat, the sealing ring is provided with a plurality of through holes, the sealing ring is connected to an opening on a side of the valve seat facing away from the valve cover, the elastic member is connected between the valve core body and the stop member, and the valve core body is abutted against the sealing ring under an action of the elastic member; and the power unit and the stop member are connected to drive the valve core body to

rotate relative to the valve seat, so that at least one channel in the valve core body and the corresponding through hole in the sealing ring are opposite to form a circulation passage for a cooling liquid.
[0007] As a possible implementation, the elastic member is a spring, the spring extends along an axial direction of the valve core body, and a first end of the spring is connected to a side of the valve core body facing away from the channel, and a second end of the spring is connected to the stop member.
[0008] As a possible implementation, the plurality of through holes have a fan-shaped cross-section; the channel is provided with a plurality of channel ports, shapes of the plurality of channel ports are all fan-shaped, and an area of each of the channel ports is smaller than an area of the through hole.
[0009] As a possible implementation, the plurality of through holes are distributed at intervals along a circumferential direction of the sealing ring; and part of the through holes and the channels are relatively distributed.
[0010] As a possible implementation, the valve core body includes a first end cover, a valve core housing and a second end cover, and the first end cover, the valve core housing and the second end cover are sequentially connected along an axis of the valve seat and form a second accommodating cavity.
[0011] As a possible implementation, the plurality of channels are located in the second accommodating cavity, and two ends of the plurality of channels are respectively connected to the plurality of channel ports, and the plurality of channel ports are located in the first end cover.
[0012] As a possible implementation, a center of the valve core housing is provided with a plurality of limiting parts, which are extending along an axial direction of the valve core housing, and a side of the limiting part facing away from the first end cover is provided with an opening; the stop member includes a plurality of limiting columns, which are extending along the axial direction of the valve core body, and the limiting column is inserted into the limiting part.
[0013] As a possible implementation, the stop member further includes a first housing and a transmission member, the limiting column and the transmission member are respectively connected to opposite sides of the first housing, the first housing and the second end cover are connected and form a third accommodating cavity, and the transmission member is connected to the power unit.
[0014] As a possible implementation, the elastic member is located in the third accommodating cavity, and two ends of the elastic member are respectively abutted against the second end cover and the first housing.
[0015] As a possible implementation, the power unit includes a second housing, a transmission gear unit and a motor, the transmission gear unit and the motor are connected to the second housing, the second housing is connected to the valve seat assembly, and the transmission gear unit is connected to the stop member.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0016] To describe the technical solutions in embodiments of the present disclosure or in the prior art more clearly, the following briefly introduces the accompanying drawings needed for describing the embodiments or the prior art. Apparently, the accompanying drawings in the following description illustrate some embodiments of the present disclosure, and persons of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative effort.
[0017] FIG. **1** is a structural explosion diagram of a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.
[0018] FIG. **2** is a structural explosion diagram of a valve core assembly and a sealing unit in a

spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.
[0019] FIG. **3** is a structural schematic diagram of a valve core body in a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.
[0020] FIG. **4** is a schematic diagram of a transmission structure of a power unit and a valve core assembly in a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.
[0021] FIG. **5** is a schematic diagram of a distribution of through holes of a sealing ring in a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.
DESCRIPTION OF REFERENCE NUMERALS
[0022] **100**—spring-loaded multi-channel electromagnetic valve; **110**—valve core assembly; **111**—valve core body; **112**—stop member; **1111**—valve core housing; **1112**—first end cover; **1113**—second end cover; **1114**—limiting part; **1115**—second accommodating cavity; **1116**—channel port; **1117**—second end cover opening; **1121**—limiting column; **1122**—first housing; **1123**—transmission member; [0023] **1116***a*—first channel port; **1116***b*—second channel port; **1116***c*—third channel port; **1116***d*—fourth channel port; **1116***e*—fifth channel port; **1116***f*—sixth channel port; **1116***g*—seventh channel port; **1116***h*—eighth channel port; [0024] **120**—valve seat assembly; **121**—valve seat; **122**—valve cover; [0025] **1211**—first accommodating cavity; **1212**—bolt connection part; [0026] **130**—sealing unit; **131**—sealing ring; **132**—elastic member; **1311**—through hole; **1321**—spring; [0027] **1311***a*—first through hole; **1311***b*—second through hole; **1311***c*—third through hole; **1311***d*—fourth through hole; **1311***e*—fifth through hole; **1311***f*—sixth through hole; **1311***g*—seventh through hole; **1311***h*—eighth through hole; **1311***i*—ninth through hole; [0028] **140**—power unit; **141**—second housing; **142**—motor; **143**—transmission gear unit; [0029] **1431**—worm gear; **1432**—worm wheel; **1433**—first gear; **1434**—second gear; **1435**—third gear; **1436**—fourth gear; **1437**—fifth gear; **150**—external sealing ring.
DESCRIPTION OF EMBODIMENTS
[0030] To make the objectives, technical solutions, and advantages of embodiments of the present disclosure clearer, the following clearly and comprehensively describes the technical solutions in embodiments of the present disclosure with reference to the accompanying drawings in embodiments of the present disclosure. Apparently, the described embodiments are merely a part rather than all embodiments of the present disclosure. All other embodiments obtained by persons of ordinary skill in the art based on embodiments of the present disclosure without creative effort shall fall within the protection scope of the present disclosure. In the absence of conflict, embodiments and features in the embodiments described below may be combined with each other.
[0031] In the prior art, a temperature control system inside the pure electric vehicles is interconnected through electromagnetic valves to achieve temperature control for the entire vehicle. The electromagnetic valves include multi-channel electromagnetic valves and integral electromagnetic valves. The multi-channel electromagnetic valves require connection via pipelines, while each integral electromagnetic valve includes a valve core and a valve body, forming a passage for a cooling liquid through rotational cooperation of the valve core and the valve body. However, during long-term use, these two types of electromagnetic valves are prone to pipeline aging and wear between the valve core and the valve body, which can lead to a reduction in the overall sealing performance of the electromagnetic valves, thereby affecting the temperature control effect of the vehicle.
[0032] In order to overcome the defects in the prior art, the present disclosure provides a spring-loaded multi-channel electromagnetic valve, including: a valve core assembly, a valve seat assembly, a sealing unit, and a power unit; where the valve seat assembly includes a valve seat and a valve cover, the valve seat is a rotary member, two ends of the valve seat are respectively provided with two openings to form a first accommodating cavity, the valve cover covers one of the two openings, and each of the valve core assembly and the sealing unit is located in the first

accommodating cavity; the valve core assembly includes a valve core body and a stop member, a shape of an outer wall of the valve core body is consistent with a shape of the opening, the valve core body is provided with a plurality of channels extending along an axial direction of the valve seat, the plurality of channels are distributed at intervals along a circumferential direction of the valve core body; the stop member is connected to the valve core body; the sealing unit includes a sealing ring and an elastic member, a shape of the sealing ring matches the shape of the opening of the valve seat, the sealing ring is provided with a plurality of through holes, the sealing ring is connected to an opening on a side of the valve seat facing away from the valve cover, the elastic member is connected between the valve core body and the stop member, and the valve core body is abutted against the sealing ring under an action of the elastic member; and the power unit and the stop member are connected to drive the valve core body to rotate relative to the valve seat, so that at least one channel in the valve core body and the corresponding through hole in the sealing ring are opposite to form a circulation passage for a cooling liquid. The valve core and the sealing ring are kept in an abutting state under an action of an elastic force of the elastic member, so that flowing of the cooling liquid in a circumferential gap between the valve core body and the valve seat is reduced, thereby improving the sealing performance of the valve body and further enhancing the temperature control effect of the vehicle.

[0033] The contents of the present disclosure will be described in detail below with reference to the accompanying drawings, so that those skilled in the art can more clearly and thoroughly understand the contents of the present disclosure.

[0034] The present disclosure provides a spring-loaded multi-channel electromagnetic valve **100**, including: a valve core assembly **110**, a valve seat assembly **120**, a sealing unit **130**, and a power unit **140**; where the valve seat assembly **120** includes a valve seat **121** and a valve cover **122**, the valve seat **121** is a rotary member, two ends of the valve seat **121** are respectively provided with two openings to form a first accommodating cavity **1211**, the valve cover **122** covers one of the two openings, and each of the valve core assembly **110** and the sealing unit **130** is located in the first accommodating cavity **1211**; the valve core assembly **110** includes a valve core body **111** and a stop member **112**, a shape of an outer wall of the valve core body **111** is consistent with a shape of the opening, the valve core body **111** is provided with a plurality of channels extending along an axial direction of the valve seat **121**, the plurality of channels are distributed at intervals along a circumferential direction of the valve core body; the stop member **112** is connected to the valve core body **111**; the sealing unit **130** includes a sealing ring **131** and an elastic member **132**, a shape of the sealing ring **131** matches the shape of the opening of the valve seat **121**, the sealing ring **131** is provided with a plurality of through holes **1311**, the sealing ring **131** is connected to an opening on a side of the valve seat **121** facing away from the valve cover **122**, the elastic member **132** is connected between the valve core body **111** and the stop member **112**, and the valve core body **111** is abutted against the sealing ring **131** under an action of the elastic member **132**; and the power unit **140** and the stop member **112** are connected to drive the valve core body **111** to rotate relative to the valve seat **121**, so that at least one channel in the valve core body **111** and the corresponding through hole **1311** in the sealing ring **131** are opposite to form a circulation passage for a cooling liquid.

[0035] FIG. **1** is a structural explosion diagram of a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure; and FIG. **2** is a structural explosion diagram of a valve core assembly and a sealing unit in a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.

[0036] As shown in FIG. **1**, the valve seat assembly **120** in the present embodiment includes a valve seat **121** and a valve cover **122**, a cross-section of the valve seat **121** is circular, the valve seat **121** is provided with two openings respectively along two ends of the valve seat **121** in an axial direction thereof to form a first accommodating cavity **1211**, the valve cover **122** is fixedly connected to one of the openings, a valve core assembly **110** is located in the first accommodating cavity **1211**, and the valve cover **122** covers an end of the valve core assembly **110** in the axial direction of the valve seat **121**.

[0037] The valve core assembly **110** includes a valve core body **111** and a stop member **112**, the valve core body **111** and the stop member **112** are connected along the axial direction of the valve seat **121**, and shapes of outer walls of the valve core body **111** and the stop member **112** are consistent with a shape of the opening of the valve seat **121**, and the valve core body **111** is sealed in the first accommodating cavity **1211** through the valve cover **122** after the valve core body **111** and the stop member **112** are connected. As shown in FIG. **2**, the valve core body **111** has a plurality of channels (not shown in the figure) inside, the plurality of channels are distributed at intervals along the circumferential direction of the valve core body **111**, and the plurality of channels all extend along an axial direction of the valve core body **111**, and each channel communicates inside the valve core body **111**.

[0038] Still as shown in FIG. **1**, an outer wall shape of the sealing ring **131** in the sealing unit **130** is also circular, and the sealing ring **131** is fixedly connected to an opening in the valve seat **121** facing away from a connection side of the valve cover **122**, and the sealing ring **131** is provided with the plurality of through holes **1311**, the plurality of through holes **1311** are distributed along a circumferential direction of the sealing ring **131** and are opposite to the plurality of channels in the valve core body **111** during the operation of the spring-loaded multi-channel electromagnetic valve. The sealing unit **130** further includes an elastic member **132**, which is connected between the valve core body **111** and the stop member **112**, and the valve core body **111** is abutted against the sealing ring **131** under an action of an elastic force of the elastic member **132**.

[0039] In the present embodiment, the power unit **140** is connected to the stop member **112**, and the power unit **140** may provide power to cause the valve core body **111** to rotate coaxially relative to the valve seat **121**. At this time, at least one channel in the valve core body **111** and the corresponding through hole **1311** in the sealing ring **131** are opposite to form a circulation passage for a cooling liquid.

[0040] According to the above configuration, while ensuring that the electromagnetic valve is provided with a plurality of circulation passages for cooling liquids, the valve core and the sealing ring are kept in an abutting state under an action of the elastic force of the elastic member, so that flowing of the cooling liquid between the valve core body and a cavity wall of the first accommodating cavity in the valve seat is reduced, thereby improving the sealing performance of the entire electromagnetic valve.

[0041] As a possible implementation, the elastic member **132** is a spring **1321**, the spring **1321** extends along an axial direction of the valve core body **111**, and a first end of the spring **1321** is connected to a side of the valve core body **111** facing away from the channel, and a second end of the spring **1321** is connected to the stop member **112**.

[0042] As shown in conjunction with FIGS. **1** and **2**, the spring **1321** extends along the axial direction of the valve core body **111**, two ends of the spring **1321** are connected to the valve core body **111** and the stop member **112**, respectively. When the stop member **112** and the valve core body **111** are connected, the spring **1321** is compressed by both the valve core body **111** and the stop member **112**, generating an elastic force that causes the spring **1321** to return to its natural charging state along its axial direction. At this time, the elastic force of the spring **1321** enables the valve core body **111** to move along the axial direction of the valve seat **121** and keep continuous abut against the sealing ring **131**, thereby improving the sealing performance among the valve core body **111**, the valve seat **121** and the sealing ring **131**.

[0043] In an implementation, the plurality of through holes **1311** have a fan-shaped cross-section; the channel is provided with a plurality of channel ports **1116**, shapes of the plurality of channel ports **1116** are all fan-shaped, and an area of each of the channel ports **1116** is smaller than an area of the through hole **1311**.

[0044] In an implementation, the plurality of through holes **1311** are distributed at intervals along a circumferential direction of the sealing ring **131**; and part of the through holes **1311** and the channels are relatively distributed.

[0045] FIG. **3** is a structural schematic diagram of a valve core body in a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure; and FIG. **5** is a schematic diagram of a distribution of through holes of a sealing ring in a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.

[0046] With reference to FIGS. **1** to **3** and **5**, the plurality of through holes **1311** in the present embodiment are located on the sealing ring **131**, and the plurality of through holes **1311** include a first through hole **1311***a*, a second through hole **1311***b*, a third through hole **1311***c*, a fourth through hole **1311***d*, a fifth through hole **1311***e*, a sixth through hole **1311***f*, a seventh through hole **1311***g*, an eighth through hole **1311***h* and a ninth through hole **1311***i*. The above through holes **1311** are sequentially distributed along the circumferential direction of the sealing ring **131**, and each through hole **1311** is fan-shaped, where areas of the first through hole **1311***a*, the second through hole **1311***b*, the third through hole **1311***c*, the fourth through hole **1311***d*, the seventh through hole **1311***g*, the eighth through hole **1311***h* and the ninth through hole **1311***i* are the same, areas of the fifth through hole **1311***e* and the sixth through hole **1311***f* are the same, and a sum of the areas of the fifth through hole **1311***e* and the sixth through hole **1311***f* is the same as the area of the other through holes.

[0047] Correspondingly, the channel port **1116** is located on a side of the valve core body **111** facing the sealing ring **131**, and the plurality of channel ports **1116** include a first channel port **1116***a*, a second channel port **1116***b*, a third channel port **1116***c*, a fourth channel port **1116***d*, a fifth channel port **1116***e*, a sixth channel port **1116***f*, a seventh channel port **1116***g* and an eighth channel port **1116***h*, and the above channel ports **1116** are distributed at intervals along a circumferential direction of the valve core body **111** on an end surface of the valve core body **111** facing the sealing ring **131**. A shape of the channel port **1116** is also fan-shaped, and a fan-shaped area of the channel port **1116** is smaller than the area of the through hole **1311**. When the valve body **111** rotates coaxially relative to the valve seat **121**, the channel port **1116** is opposite to part of the through holes **1311**.

[0048] In an implementation, as shown in FIGS. **1** and **2**, the valve core body **111** includes a first end cover **1112**, a valve core housing **1111** and a second end cover **1113**, and the first end cover **1112**, the valve core housing **1111** and the second end cover **1113** are sequentially connected along an axis of the valve seat **121** and form a second accommodating cavity **1115**. Specifically, the first end cover **1112** is welded with the valve core housing **1111**, and the valve core housing **1111** is welded with the second end cover **1113**.

[0049] As a possible implementation, the plurality of channels are located in the second accommodating cavity **1115**, and two ends of the plurality of channels are respectively connected to the plurality of channel ports **1116**, and the plurality of channel ports **1116** are located in the first end cover.

[0050] With reference to FIGS. **1-3**, the second accommodating cavity **1115** of the valve core body **111** has a plurality of channels (not shown in the figures), which are extending along the circumferential direction of the valve core body **111**, each channel has a channel port **1116**, and the channel ports **1116** are located on the first end cover **1112**. Specifically, the first channel port **1116***a* and the third channel port **1116***c* are connected by a first channel, the second channel port **1116***b* and the fourth channel port **1116***d* are connected by a second channel, the fifth channel port **1116***e* and the seventh channel port **1116***g* are connected by a third channel, the sixth channel port **1116***f* and the eighth channel port **1116***h* are connected by a fourth channel, and the cooling liquid flows in the valve core body **111** through the connections of the channels and the channel ports.

[0051] As a possible implementation, a center of the valve core housing **1111** is provided with a plurality of limiting parts **1114**, which are extending along an axial direction of the valve core housing **1111**, and a side of the limiting part **1114** facing away from the first end cover **1112** is provided with an opening; the stop member **112** includes a plurality of limiting columns **1121**, which are extending along the axial direction of the valve core body **1111**, and the limiting column **1121** is inserted into the limiting part **1114**.

[0052] As shown in FIGS. **2** and **3**, the limiting part **1114** is located at a center of the valve core housing **1111**, and the limiting part **1114** and an inner cavity wall of the second accommodating cavity **1115** are connected to form a plurality of cavities together. The limiting part **1114** extends along the axial direction of the valve core housing **1111**, and a central opening of the second end cover **1113** is a second end cover opening **1117** in FIG. **3**. The stop member **112** is provided with a plurality of limiting columns **1121**, which are also located at a center of the stop member **112**, and each limiting column **1121** extends along the axial direction of the valve core body **111** and is inserted into the limit part **1114**.

[0053] There is a possible implementation, the stop member **112** further includes a first housing **1122** and a transmission member **1123**, the limiting column **1121** and the transmission member **1123** are respectively connected to opposite sides of the first housing **1122**, the first housing **1122** and the second end cover **1113** are connected and form a third accommodating cavity, and the transmission member **1123** is connected to the power unit **140**.

[0054] Still as shown in FIG. **2**, the stop member **112** includes a limiting column **1121**, a first housing **1122**, and a transmission member **1123**, which are sequentially connected along the axial direction of the valve core body **111**, the limiting column **1121** is connected to a side of the first housing **1122** facing the valve core body **111**, and the transmission member **1123** is connected to a side of the first housing **1122** facing away from the valve core body **111**, and the first housing **1122** and the second end cover **1113** of the valve core body **111** are connected to form a third accommodating cavity (not shown in the figure).

[0055] Specifically, the limiting column **1121** of the stop member **112** passes through the second end cover opening **1117**, and is connected to the limiting part **1114** in the valve core body **111** correspondingly. The transmission member **1123** is connected to the power unit **140**, and a rotation output by the power unit **140** drives the stop member **112** to rotate, and at the same time drives the valve core body **111** to rotate. At this time, each channel port **1116** on the first end cover **1112** and the through hole **1311** on the seal ring **131** are opposite to each other, and the cooling liquid circulates through the corresponding through hole **1311** and the channel port **1116**. It should be noted that, the number of both the limiting columns **1121** and the limiting parts **1114** provided in the present embodiment is four, and they are mutually correspondingly connected. It is not difficult to understand that the number of the limiting columns **1121** may also be two, while the number of the limiting parts **1114** remains four, and the two limiting columns **1121** may be correspondingly connected to two of the limiting parts **1114**, which is still capable of achieving the connection function. Therefore, the present embodiment does not specifically limit the number of the limiting columns and the limiting parts or their relative connection position, as long as their connection satisfies the structural design requirements.

[0056] It can be understood that, the elastic member **132** is located in the third accommodating cavity, and two ends of the elastic member **132** are respectively abutted against the second end cover **1113** and the first housing **1122**. A first end of the elastic member **132** is connected to the second end cover **1113**, and a second end of the elastic member **132** is connected to the first housing **1122** and is connected to a side of the first housing **1122** facing the second end cover **1113**. When the first housing **1122** and the second end cover **1113** are connected, an elastic force of the elastic member **132** can keep the first end cover **1112** of the valve body **111** and the seal ring **131** in an abutting state.

[0057] As a possible implementation, the power unit **140** includes a second housing **141**, a transmission gear unit **143** and a motor **142**, the transmission gear unit **143** and the motor **142** are connected to the second housing **141**, the second housing **141** is connected to the valve seat assembly **120**, and the transmission gear unit **143** is connected to the stop member **112**.

[0058] FIG. **4** is a schematic diagram of a transmission structure of a power unit and a valve core assembly in a spring-loaded multi-channel electromagnetic valve according to an embodiment of the present disclosure.
[0059] With reference to FIGS. **1** and **4**, in the present embodiment, the power unit **140** includes a second housing **141**, a transmission gear unit **143**, and a motor **142**. The second housing **141** covers a side of the transmission gear unit **143** and the motor **142**, where the motor **142** is connected to the transmission gear unit **143**, and the transmission gear unit **143** is connected to the stop member **112**.
[0060] Specifically, the transmission gear unit **143** includes a worm gear **1431**, a worm wheel **1432**, a first gear **1433**, a second gear **1434**, a third gear **1435**, a fourth gear **1436** and a fifth gear **1437**. The worm gear **1431** is connected to the motor **142**, and the worm wheel **1432** is engaged with the worm gear **1431** for transmission. One side of the worm wheel **1432** is meshed with the worm gear **1431** for transmission, while the other side is meshed with the first gear **1433**. The second gear **1434** is coaxially connected to the first gear **1433**, and the second gear **1434** is meshed with the fourth gear **1436** for transmission. The third gear **1435** is coaxially connected to the fourth gear **1436**, and the third gear **1435** is meshed with the fifth gear **1437** for transmission. The fifth gear **1437** is fixedly and coaxially connected to the transmission member **1123** of the stop member **112**.
[0061] The second gear **1434** and the first gear **1433** are coaxially connected along the axial direction of the valve core body **111**, and the second gear **1434** is connected to the side facing away from the valve core body **111**; the third gear **1435** and the fourth gear **1436** are coaxially connected along the axial direction of the valve core body **111**, and the fourth gear **1436** is connected to the side facing away from the valve core body **111**. It should be noted that design parameters of the worm wheel, worm gear and gears in the transmission gear unit **143** are specifically designed according to the actual situation, and the present embodiment is not specifically limited thereto.
[0062] Through the configuration of the power unit **140**, a rotational torque output by the motor **142** can drive the valve core body **111** to rotate relative to the valve seat **121** through the transmission of the transmission gear unit **143** and the connection with the stop member **112**.
[0063] It is not difficult to understand that in the present embodiment, the valve seat **121** further has a plurality of bolt connection parts **1212**, the valve seat **121** may be connected to an external mechanism by the bolt connection parts **1212**, and the second housing **141** and the valve seat **121** may also be fixedly connected by the bolt connection parts **1212**. It should be noted that when the valve seat **121** is connected to the external mechanism, it is also necessary to seal, and specifically, it can be connected by the external sealing ring **150**. The structures of the external sealing ring **150** and the sealing ring **131** are completely the same, and when the external sealing ring **150** and the valve seat **121** are connected, through holes in the external sealing ring **150** and the through holes in the sealing ring **131** need to correspond to each other.
[0064] Hereinafter, the operating state of the spring-loaded multi-channel electromagnetic valve provided by the present embodiment will be described.
[0065] The sealing ring **131** and the valve seat **121** are fixedly connected, the valve core body **111** rotates coaxially relative to the valve seat **121**, and the channel port **1116** on the first end cover **1112** of the valve core body **111** and the through holes on the sealing ring **131** can be relatively communicated during the rotation of the valve core body **111**. In a certain operating state, the first channel port **1116***a* and the ninth through hole **1311***i* are opposite to each other, and the third channel port **1116***c* communicating with the first channel port **1116***a* through the first channel and the second through hole **1311***b* are opposite to each other. It is not difficult to understand that the cooling liquid flows through the through hole and the channel port opposite to each other as well as the channel, and it should be noted that the flow direction of the cooling liquid is not specifically limited thereto.
[0066] Correspondingly, the second channel port **1116***b* and the first through hole **1311***a* are opposite to each other, and the fourth channel port **1116***d* communicating with the second channel

port **1116***b* through the second channel and the third through hole **1311***c* are opposite to each other, and the cooling liquid flows through the through hole and the channel port opposite to each other as well as the channel.

[0067] The fifth channel port **1116***e* and the fourth through-hole **1311***d* are opposite to each other, and the seventh channel port **1116***g* communicating with the fifth channel port **1116***e* through the third channel and the seventh through hole **1311***g* are opposite to each other, and the cooling liquid flows through the through hole and the channel port opposite to each other as well as the channel.

[0068] The sixth channel port **1116***f* and the sixth through-hole **1311***f* are opposite to each other, and the eighth channel port **1116***h* communicating with the sixth channel port **1116***f* through the fourth channel and the eighth through hole **1311***h* are opposite to each other, and the cooling liquid flows through the through hole and the channel port opposite to each other as well as the channel.

[0069] The fifth through hole **1311***e* is not opposite to any of the channel ports **1116**, and at this time, the first end cover **1112** and the fifth through hole **1311***e* are opposite to each other, and the cooling liquid does not flow at the fifth through hole **1311***e*.

[0070] It can be understood that the above operating state is only one of various operating states of the spring-loaded multi-channel electromagnetic valve **100** provided by the present disclosure. The power unit **140** controls the valve core body **111** to rotate relative to the valve seat **121** and the sealing ring **131**. Correspondingly, the channel ports **1116** and the through holes **1311** may experience relative positional changes. When the valve core body **111** rotates for different angles, the channel ports **1116** are opposite to different through holes **1311**, resulting in multiple different communicating positions. This achieves the various operating states of the spring-loaded multi-channel electromagnetic valve **100**.

[0071] The present disclosure provides a spring-loaded multi-channel electromagnetic valve, including: a valve core assembly, a valve seat assembly, a sealing unit, and a power unit; where the valve seat assembly includes a valve seat and a valve cover, the valve seat is a rotary member, two ends of the valve seat are respectively provided with two openings to form a first accommodating cavity, the valve cover covers one of the two openings, and each of the valve core assembly and the sealing unit is located in the first accommodating cavity; the valve core assembly includes a valve core body and a stop member, a shape of an outer wall of the valve core body is consistent with a shape of the opening, the valve core body is provided with a plurality of channels extending along an axial direction of the valve seat, the plurality of channels are distributed at intervals along a circumferential direction of the valve core body; the stop member is connected to the valve core body; the sealing unit includes a sealing ring and an elastic member, a shape of the sealing ring matches the shape of the opening of the valve seat, the sealing ring is provided with a plurality of through holes, the sealing ring is connected to an opening on a side of the valve seat facing away from the valve cover, the elastic member is connected between the valve core body and the stop member, and the valve core body is abutted against the sealing ring under an action of the elastic member; and the power unit and the stop member are connected to drive the valve core body to rotate relative to the valve seat, so that at least one channel in the valve core body and the corresponding through hole in the sealing ring are opposite to form a circulation passage for a cooling liquid. The valve core and the sealing ring are kept in an abutting state under an action of an elastic force of the elastic member, so that flowing of the cooling liquid in a circumferential gap between the valve core body and the valve seat is reduced, thereby improving the sealing performance of the valve body and further enhancing the temperature control effect of the vehicle.

[0072] It should be noted that the terms “one embodiment,” “an embodiment,” “an exemplary embodiment,” “some embodiments,” etc., as mentioned in the specification, indicate that the described embodiments may include certain features, structures, or characteristics, but not every embodiment necessarily includes those specific features, structures, or characteristics. Moreover, such phrases do not necessarily refer to the same embodiment. Furthermore, when describing specific features, structures, or characteristics in conjunction with one embodiment, it is within the

knowledge of a person skilled in the art to implement such features, structures, or characteristics in conjunction with other embodiments, whether explicitly or implicitly described.
[0073] In general, the terms used herein should be understood at least in part based on the context of their use. For example, at least in part depending on the context, the term “one or more” as used herein may describe any feature, structure, or characteristic in the singular sense, or may describe a combination of features, structures, or characteristics in the plural sense. Similarly, at least in part depending on the context, terms such as “a” or “the” may be understood to convey singular usage or plural usage.
[0074] It should be readily understood that the terms “on,” “over,” and “above” as used in the present disclosure should be interpreted in the broadest manner possible, such that “on” not only means “directly on something,” but also includes “on something” with intermediate features or layers, and “over” or “above” includes not only the meaning of “over or above something,” but also the meaning of “over or above something” without intermediate features or layers (i.e., directly on something).
[0075] Additionally, for ease of explanation, spatial relative terms such as “below,” “beneath,” “under,” “above,” “over,” etc., may be used herein to describe the relationship of one element or feature to another element or feature as illustrated in the figures. The spatial relative terms are intended to encompass orientations other than those depicted in the figures during use or operation of the device. The apparatus may have other orientations (rotated 90 degrees or otherwise), and the spatial relative descriptive terms used herein may be interpreted accordingly.
[0076] Finally, it should be noted that the foregoing embodiments are merely intended for describing the technical solutions of the present disclosure other than limiting the present disclosure. Although the present disclosure is described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should understand that they may still make modifications to the technical solutions described in the foregoing embodiments or make equivalent substitutions to some technical features thereof. However, these modifications or substitutions do not cause the essence of the corresponding technical solutions to deviate from the scope of the technical solutions of each embodiment of the present disclosure.

Claims

1. A spring-loaded multi-channel electromagnetic valve, comprising: a valve core assembly, a valve seat assembly, a sealing unit, and a power unit, wherein the valve seat assembly comprises a valve seat and a valve cover, the valve seat is a rotary member, two ends of the valve seat are respectively provided with two openings to form a first accommodating cavity, the valve cover covers one of the two openings, and each of the valve core assembly and the sealing unit is located in the first accommodating cavity; the valve core assembly comprises a valve core body and a stop member, a shape of an outer wall of the valve core body is consistent with a shape of the opening, the valve core body is provided with a plurality of channels extending along an axial direction of the valve seat, the plurality of channels are distributed at intervals along a circumferential direction of the valve core body; the stop member is connected to the valve core body; the sealing unit comprises a sealing ring and an elastic member, a shape of the sealing ring matches the shape of the opening of the valve seat, the sealing ring is provided with a plurality of through holes, the sealing ring is connected to an opening on a side of the valve seat facing away from the valve cover, the elastic member is connected between the valve core body and the stop member, and the valve core body is abutted against the sealing ring under an action of the elastic member; and the power unit and the stop member are connected to drive the valve core body to rotate relative to the valve seat, so that at least one channel in the valve core body and the corresponding through hole in the sealing ring are opposite to form a circulation passage for a cooling liquid.
2. The spring-loaded multi-channel electromagnetic valve according to claim 1, wherein the elastic

member is a spring, the spring extends along an axial direction of the valve core body, and a first end of the spring is connected to a side of the valve core body facing away from the channel, and a second end of the spring is connected to the stop member.

3. The spring-loaded multi-channel electromagnetic valve according to claim 1, wherein the plurality of through holes have a fan-shaped cross-section; and the channel is provided with a plurality of channel ports, shapes of the plurality of channel ports are all fan-shaped, and an area of each channel port is smaller than an area of the through hole.

4. The spring-loaded multi-channel electromagnetic valve according to claim 3, wherein the plurality of through holes are distributed at intervals along a circumferential direction of the sealing ring; and parts of the through holes and the channels are relatively distributed.

5. The spring-loaded multi-channel electromagnetic valve according to claim 4, wherein the valve core body comprises a first end cover, a valve core housing, and a second end cover, and the first end cover, the valve core housing, and the second end cover are sequentially connected along an axis of the valve seat and form a second accommodating cavity.

6. The spring-loaded multi-channel electromagnetic valve according to claim 5, wherein the plurality of channels are located in the second accommodating cavity, and two ends of the plurality of channels are respectively connected to the plurality of channel ports, and the plurality of channel ports are located in the first end cover.

7. The spring-loaded multi-channel electromagnetic valve according to claim 6, wherein a center of the valve core housing is provided with a plurality of limiting parts that extend along an axial direction of the valve core housing, and a side of the limiting part facing away from the first end cover is provided with an opening; and the stop member comprises a plurality of limiting columns that extend along the axial direction of the valve core body, and the limiting column is inserted into the limiting part.

8. The spring-loaded multi-channel electromagnetic valve according to claim 7, wherein the stop member further comprises a first housing and a transmission member, the limiting column and the transmission member are respectively connected to opposite sides of the first housing, the first housing and the second end cover are connected and form a third accommodating cavity, and the transmission member is connected to the power unit.

9. The spring-loaded multi-channel electromagnetic valve according to claim 8, wherein the elastic member is located in the third accommodating cavity, and two ends of the elastic member are respectively abutted against the second end cover and the first housing.

10. The spring-loaded multi-channel electromagnetic valve according to claim 1, wherein the power unit comprises a second housing, a transmission gear unit, and a motor, the transmission gear unit and the motor are connected to the second housing, the second housing is connected to the valve seat assembly, and the transmission gear unit is connected to the stop member.

11. The spring-loaded multi-channel electromagnetic valve according to claim 2, wherein the plurality of through holes have a fan-shaped cross-section; and the channel is provided with a plurality of channel ports, shapes of the plurality of channel ports are all fan-shaped, and an area of each channel port is smaller than an area of the through hole.

12. The spring-loaded multi-channel electromagnetic valve according to claim 11, wherein the plurality of through holes are distributed at intervals along a circumferential direction of the sealing ring; and parts of the through holes and the channels are relatively distributed.

13. The spring-loaded multi-channel electromagnetic valve according to claim 12, wherein the valve core body comprises a first end cover, a valve core housing, and a second end cover, and the first end cover, the valve core housing and the second end cover are sequentially connected along an axis of the valve seat and form a second accommodating cavity.

14. The spring-loaded multi-channel electromagnetic valve according to claim 13, wherein the plurality of channels are located in the second accommodating cavity, and two ends of the plurality of channels are respectively connected to the plurality of channel ports, and the plurality of channel

ports are located in the first end cover.
15. The spring-loaded multi-channel electromagnetic valve according to claim 14, wherein a center of the valve core housing is provided with a plurality of limiting parts that extend along an axial direction of the valve core housing, and a side of the limiting part facing away from the first end cover is provided with an opening; and the stop member comprises a plurality of limiting columns that extend along the axial direction of the valve core body, and the limiting column is inserted into the limiting part.
16. The spring-loaded multi-channel electromagnetic valve according to claim 15, wherein the stop member further comprises a first housing and a transmission member, the limiting column and the transmission member are respectively connected to opposite sides of the first housing, the first housing and the second end cover are connected and form a third accommodating cavity, and the transmission member is connected to the power unit.
17. The spring-loaded multi-channel electromagnetic valve according to claim 16, wherein the elastic member is located in the third accommodating cavity, and two ends of the elastic member are respectively abutted against the second end cover and the first housing.
18. The spring-loaded multi-channel electromagnetic valve according to claim 2, wherein the power unit comprises a second housing, a transmission gear unit, and a motor, the transmission gear unit and the motor are connected to the second housing, the second housing is connected to the valve seat assembly, and the transmission gear unit is connected to the stop member.
19. The spring-loaded multi-channel electromagnetic valve according to claim 3, wherein the power unit comprises a second housing, a transmission gear unit, and a motor, the transmission gear unit and the motor are connected to the second housing, the second housing is connected to the valve seat assembly, and the transmission gear unit is connected to the stop member.
20. The spring-loaded multi-channel electromagnetic valve according to claim 4, wherein the power unit comprises a second housing, a transmission gear unit, and a motor, the transmission gear unit and the motor are connected to the second housing, the second housing is connected to the valve seat assembly, and the transmission gear unit is connected to the stop member.